

James Dwyer

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PROFESSIONAL EXPERIENCE

Union College Robotics Crew

Oct 2023 - Present

Electrical Engineering Team Lead

Schenectady, NY

- Led electrical integration and testing for Micromouse robots, coordinating PCB, motor, sensor, and power-system changes across hardware iterations to support reliable board bring-up.
- Designed custom PCBs in KiCad and Altium, integrating power distribution, motor drivers, and sensor networks into autonomous robots with iterative hardware validation.
- Collaborated with software and mechanical teams using Python and embedded C/C++ to debug control algorithms and validate sensors, advancing AI-enabled maze navigation for MIT Micromouse competition robots.

Free Flow Medical

Jun 2026 - Sep 2026

R&D Manufacturing Engineering Intern

Fremont, CA

- Integrated a three-axis CNC control system with closed-loop stepper drives, power distribution, and relay safety circuitry, supporting electrical systems testing and operational reliability.
- Developed and validated CNC manufacturing-process improvements through control-system contributions and structured process testing, strengthening production efficiency and process consistency.
- Designed and iterated SolidWorks CAD models for custom tooling, fixtures, and manufacturing components supporting CNC machining and production processes.

Avisi Technologies

Mar 2026 - May 2026

R&D Intern

Redwood City, CA

- Analyzed aqueous humor flow under varying occlusion conditions through simulation-driven evaluation, strengthening device performance validation and informing surgical design decisions in R&D.
- Prototyped a fiber optic camera system integrated with surgical forceps, enabling real-time internal visualization and advancing multidisciplinary hardware development for procedural applications.
- Executed mechanical simulations on the VisiPlate system to assess structural integrity, identify design risks, and guide performance optimization for surgical applications.

PROJECTS & OUTSIDE EXPERIENCE

Embedded Systems Developer - Residential Elevator Security System

Jan 2025 - Mar 2025

- Developed C++ firmware on an STM32 Nucleo board for a residential elevator security prototype, integrating a keypad, LCD, and motor driver to restrict access to authorized users.
- Architected modular C/C++ input-handling, motor actuation, and display feedback software using object-oriented principles, supporting reliable and scalable system operation.
- Troubleshoot driver circuits using multimeters and oscilloscopes, testing lock mechanisms under load to validate safe operation in real-world conditions.

Robotics Project Lead

Sep 2025 - Nov 2025

- Developed an Arduino-based robotic trash collector in C++ using Arduino IDE, integrating DC motors, motor drivers, and ultrasonic sensors for autonomous navigation supporting accessibility.
- Configured and tested motor-driver circuitry and power distribution through breadboarding, soldering, and power analysis to deliver stable performance and extended runtime.
- Integrated a Wi-Fi camera module with ESP32 firmware and developed a mobile app interface for remote monitoring and real-time control.

Custom Traffic Control Processor Designer

Oct 2024 - Nov 2024

- Designed a custom traffic control processor in Quartus Prime CAD software, integrating a datapath, FSM, and controller within an HLSM to simulate real-world intersection management.
- Implemented counters, comparators, and logic gates to enable real-time state transitions based on vehicle counts and timing thresholds.
- Applied digital system design techniques and FPGA-based simulation to verify scalable processor architectures for embedded applications.

EDUCATION

Union College

2027

Bachelor's, Electrical Engineering

SKILLS

Skills: Altium, KiCad, PCB Design, C/C++, STM32, ESP32, SPI / I2C / UART, Schematics, Oscilloscope, Multimeter Testing, Soldering, Power Distribution